

RVHS H2 Maths 2020 Prelim Paper 1 (Solutions)

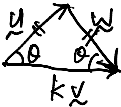
1.

$f(x) = \ln(\cos x)$ $= \ln\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots\right)$ $= \left(-\frac{x^2}{2!} + \frac{x^4}{4!}\right) - \frac{1}{2}\left(-\frac{x^2}{2!} + \frac{x^4}{4!}\right)^2 + \dots$ $= -\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{1}{2}\left(\frac{x^4}{4} + 2\left(-\frac{x^2}{2!}\right)\left(\frac{x^4}{4!}\right) + \left(\frac{x^4}{4!}\right)^2\right) + \dots$ $\approx -\frac{x^2}{2} - \frac{x^4}{12}$	
Since $\frac{d}{dx} \ln(\cos x) = \frac{-\sin x}{\cos x} = -\tan x$, $\tan x = -\frac{d}{dx} \ln(\cos x)$ $= -\frac{d}{dx} \left(-\frac{x^2}{2} - \frac{x^4}{12} + \dots\right)$ $= -\left(-x - \frac{x^3}{3}\right) + \dots$ $\approx x + \frac{x^3}{3}$	

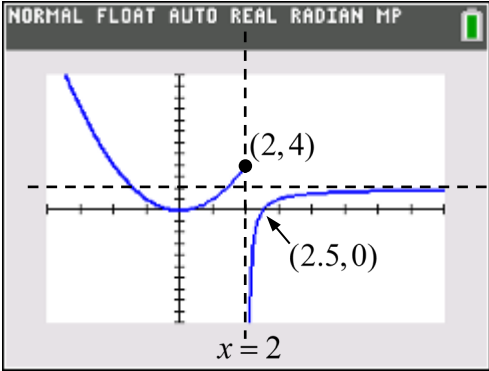
2. (a)

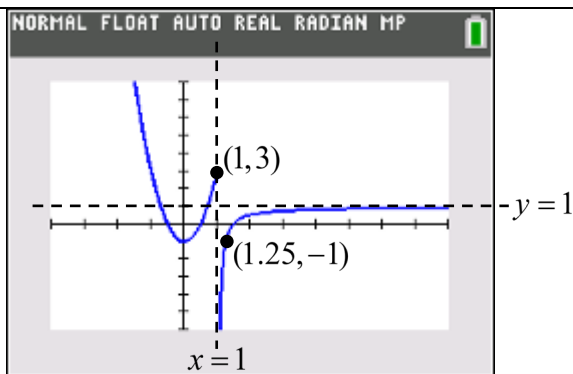
(a)	
(b) Maximal solution set = $\{x \in \mathbb{R} : x < 1\}$, for $-1 \leq a \leq 1$.	

3.

(i) $(\mathbf{u} \cdot \mathbf{v} + \mathbf{u} \times \mathbf{v})^2$ $= (\mathbf{u} \mathbf{v} \sin\theta + \mathbf{u} \mathbf{v} \cos\theta)^2$ $= (\sin\theta + \cos\theta)^2$ $= \cos^2\theta + 2\cos\theta\sin\theta + \sin^2\theta$ $= 1 + \sin 2\theta$	
(ii) $(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} = k\mathbf{v} \cdot \mathbf{v} = k \mathbf{v} ^2$ $(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} = k$ $\mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v} = k$ $\cos\theta + \cos\theta = k$ $k = 2\cos\theta$	
(iii) Since $0 < \theta < 90^\circ$, $0 < k < 2$	

4.

(i) Scale parallel to x-axis by a factor of $\frac{1}{2}$. Translate 1 unit in the negative y-direction. (OR in reverse sequence.)	
(ii) $y = f(x)$:  $y = f(2x) - 1$:	



(iii)

For the graph of $y = f'(x)$,

1. there should be a horizontal asymptote $y = 0$ instead of $y = 2$,

OR

2. it should end at $(2, 4)$ for the segment for which $x \leq 2$ since

$$\left. \frac{dy}{dx} \right|_{x=2} = 2(2) = 4, \text{ **OR**}$$

3. the segment for which $x \leq 2$ should be a straight line (with equation $y = 2x$).

(Any 2 reasons)

5.

(a)

$$\begin{aligned} \int y \, dx &= \int (\sec t + \sin t) \frac{dx}{dt} \, dt \\ &= \int (\sec t + \sin t)(-\sin t) \, dt \\ &= -\int (\tan t + \sin^2 t) \, dt \\ &= -\int \frac{\sin t}{\cos t} + \frac{(1 - \cos(2t))}{2} \, dt \\ &= \ln|\cos t| - \frac{t}{2} + \frac{1}{4} \sin(2t) + c \\ &= \ln|\cos t| - \frac{t}{2} + \frac{1}{2} \sin t \cos t + c \\ &= \ln|x| - \frac{1}{2} \cos^{-1} x + \frac{x}{2} \sqrt{1-x^2} + c \end{aligned}$$

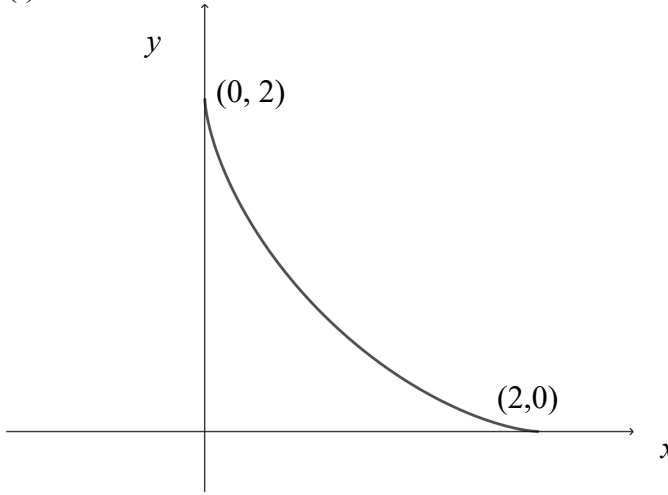
Alternatively,

$$\cos t = x \Rightarrow \sin t = \sqrt{1-x^2}$$

$$\therefore y = \frac{1}{x} + \sqrt{1-x^2}$$

$\begin{aligned}\int \sqrt{1-x^2} \, dx &= x\sqrt{1-x^2} - \int x \cdot \frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x) \, dx \\ &= x\sqrt{1-x^2} - \int \frac{-x^2}{\sqrt{1-x^2}} \, dx \\ &= x\sqrt{1-x^2} - \int \frac{1-x^2}{\sqrt{1-x^2}} \, dx - \int \frac{-1}{\sqrt{1-x^2}} \, dx \\ &= x\sqrt{1-x^2} - \int \sqrt{1-x^2} \, dx - \cos^{-1} x \\ \therefore 2 \int \sqrt{1-x^2} \, dx &= x\sqrt{1-x^2} - \cos^{-1} x \\ \Rightarrow \int \sqrt{1-x^2} \, dx &= \frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\cos^{-1} x + c \\ \text{Hence, } \int y \, dx &= \int \frac{1}{x} + \sqrt{1-x^2} \, dx \\ &= \ln x + \frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\cos^{-1} x + c\end{aligned}$	
(b) $\begin{aligned}\int_{-\sqrt{2}}^1 x^3+1 \, dx \\ &= \int_{-\sqrt{2}}^{-1} -(x^3+1) \, dx + \int_{-1}^1 x^3+1 \, dx \\ &= -\left[\frac{x^4}{4}+x\right]_{-\sqrt{2}}^{-1} + \left[\frac{x^4}{4}+x\right]_{-1}^1 \\ &= -\left(\frac{1}{4}-1-1+\sqrt{2}\right) + \left(\frac{1}{4}+1-\frac{1}{4}+1\right) \\ &= \frac{15}{4} - \sqrt{2}\end{aligned}$	

6.

(i) 	
(ii)	

$$\frac{dx}{dt} = 3 \cos^2 t (-\sin t)$$

$$\frac{dy}{dt} = 3 \sin^2 t (\cos t)$$

$$\frac{dy}{dx} = \frac{3 \sin^2 t \cos t}{-3 \sin t \cos^2 t} = -\tan t$$

P is the point $(\cos^3 p, \sin^3 p)$

Thus, the equation of the tangent:

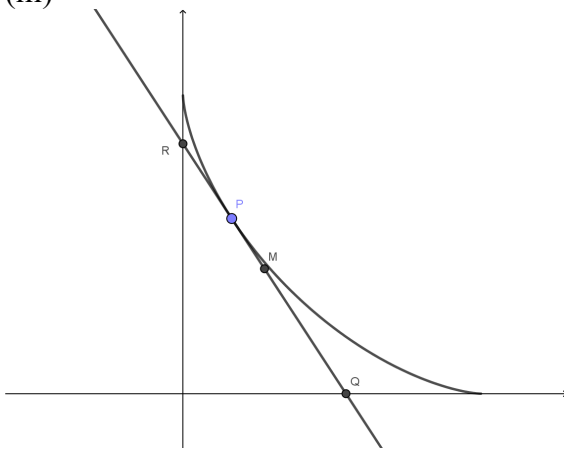
$$y - \sin^3 p = -\tan p (x - \cos^3 p)$$

$$y = (-\tan p)x + \sin p \cos^2 p + \sin^3 p$$

$$y = (-\tan p)x + \sin p (\cos^2 p + \sin^2 p)$$

$$y = (-\tan p)x + \sin p$$

(iii)



$Q(\cos p, 0), R(0, \sin p)$

Let the midpoint of QR be M . Then

$$M\left(\frac{1}{2} \cos p, \frac{1}{2} \sin p\right)$$

Therefore, parametric equations for the locus of M :

$$x = \frac{1}{2} \cos p, \quad y = \frac{1}{2} \sin p$$

Therefore, Cartesian equation for the locus of M :

$$\cos^2 p + \sin^2 p = 1$$

$$(2x)^2 + (2y)^2 = 1$$

$$x^2 + y^2 = \left(\frac{1}{2}\right)^2$$

7.

(i) $f(r) - f(r-1)$ $= \ln(3r+1) - \ln(3(r-1)+1)$ $= \ln(3r+1) - \ln(3r-2)$ $= \ln\left(\frac{3r+1}{3r-2}\right) \text{ (shown)}$	
(ii) $\sum_{r=1}^n \ln \sqrt{\frac{3r+1}{3r-2}}$ $= \frac{1}{2} \sum_{r=1}^n \ln\left(\frac{3r+1}{3r-2}\right)$ $= \frac{1}{2} \sum_{r=1}^n [f(r) - f(r-1)]$ $= \frac{1}{2} \begin{bmatrix} \cancel{f(1)} - f(0) \\ \cancel{f(2)} - \cancel{f(1)} \\ \cancel{f(3)} - \cancel{f(2)} \\ \vdots \\ \cancel{f(n-1)} - \cancel{f(n-2)} \\ f(n) - \cancel{f(n-1)} \end{bmatrix}$ $= \frac{1}{2} [f(n) - f(0)]$ $= \frac{1}{2} [\ln(3n+1) - \ln(3(0)+1)]$ $= \frac{1}{2} \ln(3n+1)$	
(iii) $\sum_{r=4}^{n-5} \ln \sqrt{\frac{3r+16}{3r+13}} = \sum_{r=5=n-5}^{n-5} \ln \sqrt{\frac{3(r-5)+16}{3(r-5)+13}} \quad \text{replace } r \text{ by } r-5$ $= \sum_{r=9}^n \ln \sqrt{\frac{3r+1}{3r-2}} \text{ (shown)}$ $\therefore \sum_{r=4}^{n-5} \ln \sqrt{\frac{3r+16}{3r+13}} = \sum_{r=1}^n \ln \sqrt{\frac{3r+1}{3r-2}} - \sum_{r=1}^8 \ln \sqrt{\frac{3r+1}{3r-2}}$ $= \frac{1}{2} \ln(3n+1) - \frac{1}{2} \ln(3(8)+1)$ $= \frac{1}{2} \ln\left(\frac{3n+1}{25}\right)$	

8.

(a)(i)

Let $x = 2 \sin u$. Then $\frac{dx}{du} = 2 \cos u$. Thus,

$$\int \sqrt{16 - 4x^2} \, dx = \int \sqrt{16 - 4(2^2 \sin^2 u)} (2 \cos u) \, du$$

$$= 8 \int \sqrt{1 - \sin^2 u} (\cos u) \, du$$

$$= 8 \int \cos^2 u \, du$$

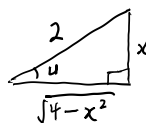
$$= 4 \int \cos 2u + 1 \, du$$

$$= 4 \left[\frac{1}{2} \sin 2u + u \right] + c$$

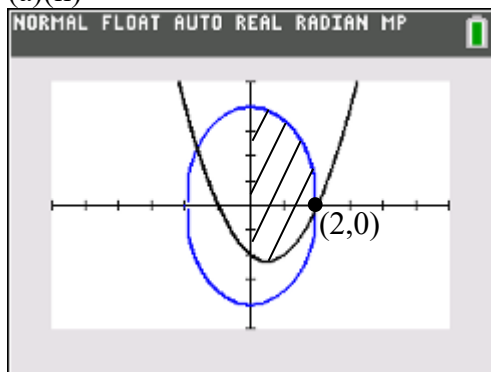
$$= 4 \sin u \cos u + 4u + c$$

$$= 2x \frac{\sqrt{4 - x^2}}{2} + 4 \sin^{-1} \left(\frac{x}{2} \right) + c$$

$$= x\sqrt{4 - x^2} + 4 \sin^{-1} \left(\frac{x}{2} \right) + c$$



(a)(ii)



Note: The curves intersect at (2,0).

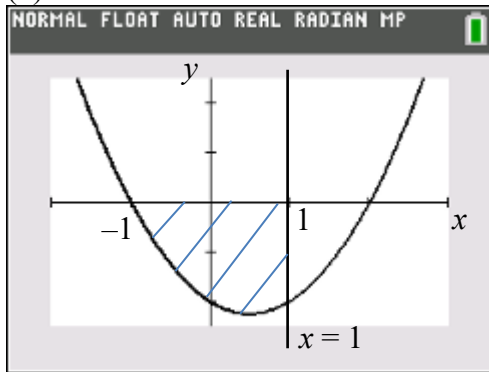
$$\text{Area} = \int_0^2 \sqrt{16 - 4x^2} - (x^2 - x - 2) \, dx$$

$$= \left[x\sqrt{4 - x^2} + 4 \sin^{-1} \left(\frac{x}{2} \right) - \frac{1}{3} x^3 + \frac{1}{2} x^2 + 2x \right]_0^2$$

$$= \left[2\pi - \frac{8}{3} + 2 + 4 \right]$$

$$= 2\pi + \frac{10}{3} \text{ units}^2$$

(b)



$$\begin{aligned}
 \text{Volume} &= \pi \int_{-1}^1 y^2 \, dx \\
 &= \pi \int_{-1}^1 (x^2 - x - 2)^2 \, dx \\
 &= 20.1 \text{ units}^3 \text{ (3 s.f.)}
 \end{aligned}$$

9.

(a)

$$z - 2w^* = i \quad (1)$$

$$iz - w = i \quad (2)$$

From (2), $iz = w + i$

$$z = 1 - wi \quad (3)$$

sub (3) into (1): $1 - wi - 2w^* = i$

let $w = a + bi$,

$$1 - (a + bi)i - 2(a - bi) = i$$

$$1 - ai + b - 2a + 2bi = i$$

$$1 - 2a + b + (2b - a)i = i$$

Comparing real part,

$$1 - 2a + b = 0$$

$$2a - b = 1 \quad (4)$$

Comparing imaginary part,

$$2b - a = 1 \quad (5)$$

Solving (4) and (5) using GC, $a = 1$, $b = 1$

Therefore, $w = 1 + i$,

$$z = 1 - (1 + i)i$$

$$= 2 - i$$

Alternate solution

$$z - 2w^* = i \quad (1)$$

$$iz - w = i \quad (2)$$

let $z = x + yi$, $w = a + bi$, From (1), $z - 2w^* = i$ $(x + yi) - 2(a - bi) = i$ $(x - 2a) + (y + 2b)i = i$ Comparing real and imaginary parts, $x - 2a = 0$ (3) $y + 2b = 1$ (4) From (2), $iz - w = i$ $i(x + yi) - (a + bi) = i$ $ix - y - a - bi = i$ $(-a - y) + (x - b)i = i$ Comparing real and imaginary parts, $-a - y = 0$ (5) $x - b = 1$ (6) Solving (3), (4), (5) and (6) using GC, $x = 2, y = -1, a = 1, b = 1$ Therefore, $w = 1 + i, z = 2 - i$	
(b)(i) $u = \sqrt{(-1)^2 + (-\sqrt{3})^2} = 2$ $\arg(u) = -\pi + \tan^{-1}\left(\frac{\sqrt{3}}{1}\right)$ $= -\frac{2\pi}{3}$	
(b)(ii) $vu^* = v u^*$ $= 2\sqrt{2}$ $\arg(vu^*) = \arg(v) + \arg(u^*)$ $= \arg(v) - \arg(u)$ $= \frac{7\pi}{12} - \left(-\frac{2\pi}{3}\right)$ $= \frac{15\pi}{12}$ $= -\frac{3\pi}{4}$ $\therefore vu^* = 2\sqrt{2}e^{i\left(-\frac{3\pi}{4}\right)}$	
(b)(iii) For $(vu^*)^n$ to be purely imaginary, $\arg((vu^*)^n) = (2k+1)\frac{\pi}{2}$ for $k \in \mathbb{Z}$	

$n \arg(vu^*) = (2k+1)\frac{\pi}{2}$ $-\frac{3n\pi}{4} = \frac{(2k+1)\pi}{2}$ $-3n = 4k + 2$ $n = \frac{-2}{3}(2k+1)$ $\therefore \text{smallest positive integer } n = 2$	
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10.

(i) June 2021 is the 13 th month. Required no. of clients $= T_3$ $= 2 + (13 - 1)(3)$ $= 38$																
(ii) Want $S_n \geq 500$ $\frac{n}{2}[2(2) + (n - 1)(3)] \geq 500$ $n(3n + 1) \geq 1000$ $3n^2 + n - 1000 \geq 0$ $n \leq -18.4248$ or $n \geq 18.0915$ Hence the agent needs 19 months to achieve a clientele of at least 500.																
(iii) <table><tr><td>Month</td><td>Clientele size before attrition</td><td>After attrition</td></tr><tr><td>1</td><td>$500 + 3$</td><td>$0.99(500 + 3)$</td></tr><tr><td>2</td><td>$0.99(500) + 0.99(3) + 3$</td><td>$(0.99)^2(500) + (0.99)^2(3) + (0.99)3$</td></tr><tr><td>3</td><td></td><td>$(0.99)^3(500) + (0.99)^3(3) + (0.99)^2(3) + (0.99)3$</td></tr><tr><td>$n$</td><td></td><td>$(0.99)^n(500) + (0.99)^n(3) + (0.99)^{n-1}(3) + \dots + (0.99)3$</td></tr></table> Size of clientele at end of n^{th} month $= (0.99)^n(500) + (0.99)^n(3) + (0.99)^{n-1}(3) + \dots + (0.99)3$ $= 500(0.99)^n + \frac{3(0.99)(1 - 0.99^n)}{1 - 0.99}$ $= 500(0.99)^n + 297(1 - 0.99^n)$ $= 297 + 203(0.99)^n$ (shown)	Month	Clientele size before attrition	After attrition	1	$500 + 3$	$0.99(500 + 3)$	2	$0.99(500) + 0.99(3) + 3$	$(0.99)^2(500) + (0.99)^2(3) + (0.99)3$	3		$(0.99)^3(500) + (0.99)^3(3) + (0.99)^2(3) + (0.99)3$	n		$(0.99)^n(500) + (0.99)^n(3) + (0.99)^{n-1}(3) + \dots + (0.99)3$	
Month	Clientele size before attrition	After attrition														
1	$500 + 3$	$0.99(500 + 3)$														
2	$0.99(500) + 0.99(3) + 3$	$(0.99)^2(500) + (0.99)^2(3) + (0.99)3$														
3		$(0.99)^3(500) + (0.99)^3(3) + (0.99)^2(3) + (0.99)3$														
n		$(0.99)^n(500) + (0.99)^n(3) + (0.99)^{n-1}(3) + \dots + (0.99)3$														
(iv) As $n \rightarrow \infty$, $203(0.99)^n \rightarrow 0$. So the clientele size converges to 297.																

(v) let %age loss be $(1 - x) \times 100\%$

Find x s.t. $\lim_{n \rightarrow \infty} [x^n(500) + x^n(3) + x^{n-1}(3) + \dots + x(3)] \geq 350$

$$\Rightarrow \lim_{n \rightarrow \infty} \left[(500)x^n + \frac{3x(1 - x^n)}{1 - x} \right] \geq 350$$

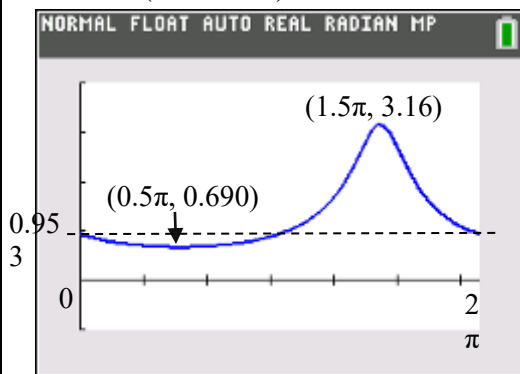
$$\Rightarrow \frac{3x}{1 - x} \geq 350 \quad \text{since } x^n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow x \geq 0.9915014$$

Hence, the maximum percentage loss that agent B can have is $(1 - 0.9915014) \times 100\% = 0.84985\% = 0.850\%$ (3 sf)

11.

(a) $y = (\sin x + 1.1)^{-0.5}$



(b)(i)

$$E = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

$$mv^2 = 2E + \frac{2GMm}{r}$$

$$v^2 = \frac{2E}{m} + \frac{2GM}{r} = \frac{GM}{12 \left(\sin \left(\frac{t}{76} \right) + 1.1 \right)^{0.5}} + \frac{2E}{m} \quad (\text{shown})$$

(b)(ii)

$$v^2 = \frac{GM}{12 \left(\sin \left(\frac{t}{76} \right) + 1.1 \right)^{0.5}} + \frac{2E}{m}$$

$$= \frac{GM}{12} \left(\sin \left(\frac{t}{76} \right) + 1.1 \right)^{-0.5} + \frac{2E}{m}$$

$$2v \left(\frac{dv}{dt} \right) = -\frac{GM}{24} \left(\sin \left(\frac{t}{76} \right) + 1.1 \right)^{-1.5} \cdot \left(\cos \left(\frac{t}{76} \right) \right) \cdot \left(\frac{1}{76} \right)$$

For stationary values,

$0 = \frac{-GM}{1824} \left(\sin\left(\frac{t}{76}\right) + 1.1 \right)^{-1.5} \cdot \left(\cos\left(\frac{t}{76}\right) \right)$ $\cos\frac{t}{76} = 0$ $t = 38\pi, 114\pi$ We don't need to find other values of t since it will have the same repeated set of speeds each time. Deducing from part (a), maximum when $t = 114\pi$ When $t = 114\pi$, $v^2 = \frac{GM}{12 \left(\sin\left(\frac{114\pi}{76}\right) + 1.1 \right)^{0.5}} + \frac{2E}{m}$ $= \frac{GM}{12 (\sin(1.5\pi) + 1.1)^{0.5}} + \frac{2E}{m}$ $v^2 = 0.263523GM + \frac{2E}{m}$ $v = \sqrt{0.264GM + \frac{2E}{m}}$	
(b)(iii) $\frac{dL}{dt} = \frac{dL}{dr} \cdot \frac{dr}{dt}$ $r = 24 \left(\sin\left(\frac{t}{76}\right) + 1.1 \right)^{0.5}$ $\frac{dr}{dt} = 12 \left(\sin\left(\frac{t}{76}\right) + 1.1 \right)^{-0.5} \left(\cos\left(\frac{t}{76}\right) \right) \left(\frac{1}{76} \right)$ $= \frac{3 \cos\left(\frac{t}{76}\right)}{19 \left(\sin\left(\frac{t}{76}\right) + 1.1 \right)^{0.5}}$ $\left. \frac{dr}{dt} \right _{t=0} = \frac{3 \cos(0)}{19 (\sin(0) + 1.1)^{0.5}} = 0.150547$ Given $\left. \frac{dL}{dr} \right _{t=0} = -2$ $\frac{dL}{dt} = (-2)(0.150547) = -0.301094$ Therefore, the luminance is decreasing at a rate of 0.301 units per year at the time of discovery.	

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